Newman-Penrose Formalism Calculations

Stephani et al., Equation (31.41_2)

Metric

$$ds^2 = -H^2 \mathrm{d}u \otimes \mathrm{d}u + \left(-\frac{l^2 - z^2}{l^2 + z^2}\right) \mathrm{d}u \otimes \mathrm{d}v + \left(\frac{l^2 vz - 2\,mvz^2 - vz^3}{l^4 + 2\,l^2 z^2 + z^4}\right) \mathrm{d}u \otimes \mathrm{d}z + \left(-\frac{l^3 v - 2\,lmvz - lvz^2}{l^4 + 2\,l^2 z^2 + z^4}\right) \mathrm{d}u \otimes \mathrm{d}y + \left(-\frac{l^2 - z^2}{l^2 + z^2}\right) \mathrm{d}v \otimes \mathrm{d}u + \left(\frac{l^2 vz - 2\,mvz^2 - vz^3}{l^4 + 2\,l^2 z^2 + z^4}\right) \mathrm{d}z \otimes \mathrm{d}u + \mathrm{d}z \otimes \mathrm{d}z + \left(-\frac{l^3 v - 2\,lmvz - lvz^2}{l^4 + 2\,l^2 z^2 + z^4}\right) \mathrm{d}y \otimes \mathrm{d}u + \left(-\frac{l^2 + z^2}{l^2 - 2\,mz - z^2}\right) \mathrm{d}y \otimes \mathrm{d}y$$

Coordinates

$(x^\mu) = (u, v, z, y)$

Metric Functions

$$H = \frac{2\,l^4 v^2}{l^6 + 3\,l^4 z^2 + 3\,l^2 z^4 + z^6} - \frac{4\,l^2 mv^2 z}{l^6 + 3\,l^4 z^2 + 3\,l^2 z^4 + z^6} - \frac{2\,l^2 v^2 z^2}{l^6 + 3\,l^4 z^2 + 3\,l^2 z^4 + z^6} - \frac{v^2}{2\,(l^2 + z^2)}$$

Null Tetrad

Covariant components

$l = (1)\ du$

$$n = \left(\frac{1}{2}\,H^2\right)\ du + \left(\frac{l^2}{l^2 + z^2} - \frac{z^2}{l^2 + z^2}\right)\ dv + \left(-\frac{l^2 vz}{l^4 + 2\,l^2 z^2 + z^4} + \frac{2\,mvz^2}{l^4 + 2\,l^2 z^2 + z^4} + \frac{vz^3}{l^4 + 2\,l^2 z^2 + z^4}\right)\ dz + \left(\frac{l^3 v}{l^4 + 2\,l^2 z^2 + z^4} - \frac{2\,lmvz}{l^4 + 2\,l^2 z^2 + z^4} - \frac{lvz^2}{l^4 + 2\,l^2 z^2 + z^4}\right)\ dy$$

$$m = \left(\sqrt{\frac{1}{2}}\right)\ dz + \left(\frac{i\,\sqrt{\frac{1}{2}}\sqrt{l^2 + z^2}}{\sqrt{-l^2 + 2\,mz + z^2}}\right)\ dy$$

$$\overline{m} = \left(\sqrt{\frac{1}{2}}\right) dz + \left(-\frac{i\sqrt{\frac{1}{2}}\sqrt{l^2+z^2}}{\sqrt{-l^2+2\,mz+z^2}}\right) dy$$

Contravariant components

$$l = \left(-\frac{l^2}{l^2-z^2} - \frac{z^2}{l^2-z^2}\right) dv$$

$$n = (-1)\, du + \left(\frac{l^2H^2}{2\,(l^2-z^2)} + \frac{z^2H^2}{2\,(l^2-z^2)}\right) dv$$

$$\begin{aligned} m = \Bigg(&\frac{i\sqrt{2}\sqrt{l^2+z^2}l^5v}{2\,(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-l^2+2\,mz+z^2}} + \frac{\sqrt{2}l^4vz}{2\,(l^6+l^4z^2-l^2z^4-z^6)} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}l^3mvz}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-l^2+2\,mz+z^2}} - \frac{i\sqrt{2}\sqrt{l^2+z^2}l^3vz^2}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-l^2+2\,mz+z^2}} \\ &- \frac{\sqrt{2}l^2mvz^2}{l^6+l^4z^2-l^2z^4-z^6} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}lm^2vz^2}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-l^2+2\,mz+z^2}} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}lmvz^3}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-l^2+2\,mz+z^2}} + \frac{i\sqrt{2}\sqrt{l^2+z^2}lvz^4}{2\,(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-l^2+2\,mz+z^2}} \\ &- \frac{\sqrt{2}mvz^4}{l^6+l^4z^2-l^2z^4-z^6} - \frac{\sqrt{2}vz^5}{2\,(l^6+l^4z^2-l^2z^4-z^6)}\Bigg) dv + \left(\sqrt{\frac{1}{2}}\right) dz + \left(-\frac{i\sqrt{2}l^2}{2\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} + \frac{i\sqrt{2}mz}{\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} + \frac{i\sqrt{2}z^2}{2\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\right) dy \end{aligned}$$

$$\begin{aligned} \overline{m} = \Bigg(&-\frac{i\sqrt{2}\sqrt{l^2+z^2}l^5v}{2\,(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-l^2+2\,mz+z^2}} + \frac{\sqrt{2}l^4vz}{2\,(l^6+l^4z^2-l^2z^4-z^6)} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}l^3mvz}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-l^2+2\,mz+z^2}} + \frac{i\sqrt{2}\sqrt{l^2+z^2}l^3vz^2}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-l^2+2\,mz+z^2}} \\ &- \frac{\sqrt{2}l^2mvz^2}{l^6+l^4z^2-l^2z^4-z^6} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}lm^2vz^2}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-l^2+2\,mz+z^2}} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}lmvz^3}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-l^2+2\,mz+z^2}} - \frac{i\sqrt{2}\sqrt{l^2+z^2}lvz^4}{2\,(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-l^2+2\,mz+z^2}} \\ &- \frac{\sqrt{2}mvz^4}{l^6+l^4z^2-l^2z^4-z^6} - \frac{\sqrt{2}vz^5}{2\,(l^6+l^4z^2-l^2z^4-z^6)}\Bigg) dv + \left(\sqrt{\frac{1}{2}}\right) dz + \left(\frac{i\sqrt{2}l^2}{2\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} - \frac{i\sqrt{2}mz}{\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} - \frac{i\sqrt{2}z^2}{2\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\right) dy \end{aligned}$$

Directional Derivatives

$$D = -\frac{l^2}{l^2-z^2} - \frac{z^2}{l^2-z^2}\,\partial_v$$

$$\Delta = -\partial_u + \frac{l^2H^2}{2\,(l^2-z^2)} + \frac{z^2H^2}{2\,(l^2-z^2)}\,\partial_v$$

$$\delta = \frac{i\sqrt{2}\sqrt{l^2+z^2}l^5v}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}} + \frac{\sqrt{2}l^4vz}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}l^3mvz}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}} - \frac{i\sqrt{2}\sqrt{l^2+z^2}l^3vz^2}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}}$$
$$- \frac{\sqrt{2}l^2mvz^2}{l^6+l^4z^2-l^2z^4-z^6} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}lm^2vz^2}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}lmvz^3}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}} + \frac{i\sqrt{2}\sqrt{l^2+z^2}lvz^4}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}}$$
$$- \frac{\sqrt{2}mvz^4}{l^6+l^4z^2-l^2z^4-z^6} - \frac{\sqrt{2}vz^5}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)}\,\partial_v + \sqrt{\frac{1}{2}}\,\partial_z - \frac{i\sqrt{2}l^2}{2\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} + \frac{i\sqrt{2}mz}{\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} + \frac{i\sqrt{2}z^2}{2\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\,\partial_y$$

$$\overline{\delta} = -\frac{i\sqrt{2}\sqrt{l^2+z^2}l^5v}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}} + \frac{\sqrt{2}l^4vz}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}l^3mvz}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}} + \frac{i\sqrt{2}\sqrt{l^2+z^2}l^3vz^2}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}}$$
$$- \frac{\sqrt{2}l^2mvz^2}{l^6+l^4z^2-l^2z^4-z^6} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}lm^2vz^2}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}lmvz^3}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}} - \frac{i\sqrt{2}\sqrt{l^2+z^2}lvz^4}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}}$$
$$- \frac{\sqrt{2}mvz^4}{l^6+l^4z^2-l^2z^4-z^6} - \frac{\sqrt{2}vz^5}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)}\,\partial_v + \sqrt{\frac{1}{2}}\,\partial_z + \frac{i\sqrt{2}l^2}{2\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} - \frac{i\sqrt{2}mz}{\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} - \frac{i\sqrt{2}z^2}{2\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\,\partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{i\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}l^5mv}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}} + \frac{i\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}l^5vz}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}} - \frac{2i\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}l^3vz^3}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}}$$
$$+ \frac{3i\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}lmvz^4}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}} + \frac{i\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}lvz^5}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}}$$

$$\lambda = 0$$

$$
\begin{aligned}
\nu = & -\frac{i\sqrt{2}l^5 vH\frac{\partial}{\partial v}H}{2\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} + \frac{2i\sqrt{2}l^3 mvzH\frac{\partial}{\partial v}H}{\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} + \frac{i\sqrt{2}l^3 vz^2H\frac{\partial}{\partial v}H}{\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} - \frac{2i\sqrt{2}lm^2vz^2H\frac{\partial}{\partial v}H}{\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} \\
& - \frac{2i\sqrt{2}lmvz^3H\frac{\partial}{\partial v}H}{\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} - \frac{i\sqrt{2}lvz^4H\frac{\partial}{\partial v}H}{2\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} + \frac{i\sqrt{2}l^5H^2}{4\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} - \frac{i\sqrt{2}l^3mzH^2}{\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} \\
& - \frac{i\sqrt{2}l^3z^2H^2}{2\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} + \frac{i\sqrt{2}lm^2z^2H^2}{\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} + \frac{i\sqrt{2}lmz^3H^2}{\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} + \frac{i\sqrt{2}lz^4H^2}{4\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} \\
& + \frac{\sqrt{2}l^4H\frac{\partial}{\partial z}H}{2\left(l^4-z^4\right)} - \frac{\sqrt{2}z^4H\frac{\partial}{\partial z}H}{2\left(l^4-z^4\right)} + \frac{\sqrt{2}l^2vzH\frac{\partial}{\partial v}H}{2\left(l^4-z^4\right)} - \frac{\sqrt{2}mvz^2H\frac{\partial}{\partial v}H}{l^4-z^4} - \frac{\sqrt{2}vz^3H\frac{\partial}{\partial v}H}{2\left(l^4-z^4\right)} + \frac{3\sqrt{2}l^2zH^2}{4\left(l^4-z^4\right)} + \frac{\sqrt{2}mz^2H^2}{2\left(l^4-z^4\right)} + \frac{\sqrt{2}z^3H^2}{4\left(l^4-z^4\right)}
\end{aligned}
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\begin{aligned}
\alpha = {} & \frac{i\sqrt{l^2+z^2}\,l^7}{4\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)\sqrt{-l^2+2\,mz+z^2}} \\
& - \frac{l^6m}{2\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)} - \frac{l^6z}{4\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)} \\
& - \frac{3i\sqrt{l^2+z^2}\,l^5mz}{2\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)\sqrt{-l^2+2\,mz+z^2}} \\
& - \frac{3i\sqrt{l^2+z^2}\,l^5z^2}{4\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)\sqrt{-l^2+2\,mz+z^2}} \\
& - \frac{l^4mz^2}{2\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)} \\
& + \frac{3i\sqrt{l^2+z^2}\,l^3m^2z^2}{\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)\sqrt{-l^2+2\,mz+z^2}} \\
& + \frac{l^4z^3}{4\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)} \\
& + \frac{3i\sqrt{l^2+z^2}\,l^3mz^3}{\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)\sqrt{-l^2+2\,mz+z^2}} \\
& - \frac{l^2m^2z^3}{\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8} \\
& - \frac{2i\sqrt{l^2+z^2}\,lm^3z^3}{\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)\sqrt{-l^2+2\,mz+z^2}} \\
& + \frac{3i\sqrt{l^2+z^2}\,l^3z^4}{4\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)\sqrt{-l^2+2\,mz+z^2}} \\
& - \frac{3\,l^2mz^4}{2\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)} \\
& - \frac{3i\sqrt{l^2+z^2}\,lm^2z^4}{\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)\sqrt{-l^2+2\,mz+z^2}} \\
& + \frac{l^2z^5}{4\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)} \\
& - \frac{3i\sqrt{l^2+z^2}\,lmz^5}{2\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)\sqrt{-l^2+2\,mz+z^2}} \\
& - \frac{m^2z^5}{\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8} \\
& - \frac{i\sqrt{l^2+z^2}\,lz^6}{4\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)\sqrt{-l^2+2\,mz+z^2}} \\
& - \frac{3\,mz^6}{2\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)} - \frac{z^7}{4\left(\sqrt{2}l^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3-2\sqrt{2}l^4z^4+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7+\sqrt{2}z^8\right)}
\end{aligned}
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\begin{aligned}
\beta = &-\frac{i\,l^7}{4\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\\
&+\frac{3i\,l^5mz}{2\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\\
&+\frac{3i\,l^5z^2}{4\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\\
&-\frac{3i\,l^3m^2z^2}{\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\\
&-\frac{3i\,l^3mz^3}{\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\\
&+\frac{2i\,lm^3z^3}{\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\\
&-\frac{3i\,l^3z^4}{4\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\\
&+\frac{3i\,lm^2z^4}{\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\\
&+\frac{3i\,lmz^5}{2\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\\
&+\frac{i\,lz^6}{4\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}+\frac{l^4m}{2\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)}\\
&+\frac{7\,l^4z}{4\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)}-\frac{2\,l^2mz^2}{\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6}\\
&-\frac{3\,l^2z^3}{2\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)}-\frac{m^2z^3}{\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6}\\
&-\frac{mz^4}{2\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)}-\frac{z^5}{4\left(\sqrt{2}l^6-2\sqrt{2}l^4mz-\sqrt{2}l^4z^2-\sqrt{2}l^2z^4+2\sqrt{2}mz^5+\sqrt{2}z^6\right)}
\end{aligned}
$$

$$
\begin{aligned}
\tau = &-\frac{i\sqrt{2}l^5}{4\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}+\frac{i\sqrt{2}l^3mz}{\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}+\frac{i\sqrt{2}l^3z^2}{2\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}-\frac{i\sqrt{2}lm^2z^2}{\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}\\
&-\frac{i\sqrt{2}lmz^3}{\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}-\frac{i\sqrt{2}lz^4}{4\left(l^4-z^4\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}+\frac{3\sqrt{2}l^2z}{4\left(l^4-z^4\right)}+\frac{\sqrt{2}mz^2}{2\left(l^4-z^4\right)}+\frac{\sqrt{2}z^3}{4\left(l^4-z^4\right)}
\end{aligned}
$$

$$\pi = -\frac{i\sqrt{2}\sqrt{l^2+z^2}l^5}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}} - \frac{3\sqrt{2}l^4z}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)} + \frac{i\sqrt{2}\sqrt{l^2+z^2}l^3mz}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}}$$
$$+ \frac{i\sqrt{2}\sqrt{l^2+z^2}l^3z^2}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}} - \frac{\sqrt{2}l^2mz^2}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} - \frac{i\sqrt{2}\sqrt{l^2+z^2}lm^2z^2}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}} - \frac{\sqrt{2}l^2z^3}{l^6+l^4z^2-l^2z^4-z^6}$$
$$- \frac{i\sqrt{2}\sqrt{l^2+z^2}lmz^3}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}} - \frac{i\sqrt{2}\sqrt{l^2+z^2}lz^4}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-l^2+2\,mz+z^2}} - \frac{\sqrt{2}mz^4}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} - \frac{\sqrt{2}z^5}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)}$$

$$\epsilon = 0$$

$$\gamma = -\frac{l^8H\frac{\partial}{\partial v}H}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)} - \frac{2\,l^6z^2H\frac{\partial}{\partial v}H}{l^8+2\,l^6z^2-2\,l^2z^6-z^8} - \frac{3\,l^4z^4H\frac{\partial}{\partial v}H}{l^8+2\,l^6z^2-2\,l^2z^6-z^8} - \frac{2\,l^2z^6H\frac{\partial}{\partial v}H}{l^8+2\,l^6z^2-2\,l^2z^6-z^8} - \frac{z^8H\frac{\partial}{\partial v}H}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)}$$
$$- \frac{i\,l^7mv}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} - \frac{i\,l^7vz}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}$$
$$+ \frac{i\,l^5m^2vz}{\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} + \frac{3i\,l^5mvz^2}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}$$
$$+ \frac{3i\,l^5vz^3}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} - \frac{7i\,l^3mvz^4}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} - \frac{3i\,l^3vz^5}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}$$
$$+ \frac{3i\,lm^2vz^5}{\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} + \frac{5i\,lmvz^6}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}} + \frac{i\,lvz^7}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{l^2+z^2}\sqrt{-l^2+2\,mz+z^2}}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$
\begin{aligned}
\Phi_{02} = {} & \frac{5\,l^{10}}{8\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} - \frac{l^{8}mz}{4\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} \\
& - \frac{11\,l^{8}z^{2}}{8\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} - \frac{11\,l^{6}m^{2}z^{2}}{2\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} \\
& - \frac{23\,l^{6}mz^{3}}{4\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} + \frac{4\,l^{4}m^{3}z^{3}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} \\
& - \frac{l^{6}z^{4}}{2\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} + \frac{7\,l^{4}m^{2}z^{4}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} \\
& - \frac{2\,l^{2}m^{4}z^{4}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} + \frac{5\,l^{4}mz^{5}}{4\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} \\
& - \frac{3\,l^{2}m^{3}z^{5}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} + \frac{5\,l^{4}z^{6}}{2\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} \\
& - \frac{l^{2}m^{2}z^{6}}{2\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} - \frac{9\,l^{2}mz^{7}}{4\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} \\
& + \frac{m^{3}z^{7}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} - \frac{9\,l^{2}z^{8}}{8\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} \\
& - \frac{m^{2}z^{8}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} - \frac{mz^{9}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} \\
& - \frac{z^{10}}{8\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} - \frac{i\sqrt{l^{2}+z^{2}}\sqrt{-l^{2}+2\,mz+z^{2}}l^{7}m}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} \\
& - \frac{9i\sqrt{l^{2}+z^{2}}\sqrt{-l^{2}+2\,mz+z^{2}}l^{7}z}{4\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} + \frac{2i\sqrt{l^{2}+z^{2}}\sqrt{-l^{2}+2\,mz+z^{2}}l^{5}m^{2}z}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} + \dots
\end{aligned}
$$

$$\Phi_{10} = 0$$

$$
\begin{aligned}
\Phi_{11} = {} & \frac{l^{14}\frac{\partial}{\partial v}H^{2}}{4\,(l^{14}-4\,l^{12}mz-l^{12}z^{2}+4\,l^{10}m^{2}z^{2}-3\,l^{10}z^{4}+4\,l^{8}m^{2}z^{4}+12\,l^{8}mz^{5}+3\,l^{8}z^{6}-8\,l^{6}m^{2}z^{6}+3\,l^{6}z^{8}-8\,l^{4}m^{2}z^{8}-12\,l^{4}mz^{9}-3\,l^{4}z^{10}+4\,l^{2}m^{2}z^{10}-l^{2}z^{12}+4\,m^{2}z^{12}+4\,mz^{13}+z^{14})} \\
& - \frac{l^{12}mz\frac{\partial}{\partial v}H^{2}}{l^{14}-4\,l^{12}mz-l^{12}z^{2}+4\,l^{10}m^{2}z^{2}-3\,l^{10}z^{4}+4\,l^{8}m^{2}z^{4}+12\,l^{8}mz^{5}+3\,l^{8}z^{6}-8\,l^{6}m^{2}z^{6}+3\,l^{6}z^{8}-8\,l^{4}m^{2}z^{8}-12\,l^{4}mz^{9}-3\,l^{4}z^{10}+4\,l^{2}m^{2}z^{10}-l^{2}z^{12}+4\,m^{2}z^{12}+4\,mz^{13}+z^{14}} \\
& + \frac{3\,l^{12}z^{2}\frac{\partial}{\partial v}H^{2}}{4\,(l^{14}-4\,l^{12}mz-l^{12}z^{2}+4\,l^{10}m^{2}z^{2}-3\,l^{10}z^{4}+4\,l^{8}m^{2}z^{4}+12\,l^{8}mz^{5}+3\,l^{8}z^{6}-8\,l^{6}m^{2}z^{6}+3\,l^{6}z^{8}-8\,l^{4}m^{2}z^{8}-12\,l^{4}mz^{9}-3\,l^{4}z^{10}+4\,l^{2}m^{2}z^{10}-l^{2}z^{12}+4\,m^{2}z^{12}+4\,mz^{13}+z^{14})} \\
& + \frac{l^{10}m^{2}z^{2}\frac{\partial}{\partial v}H^{2}}{l^{14}-4\,l^{12}mz-l^{12}z^{2}+4\,l^{10}m^{2}z^{2}-3\,l^{10}z^{4}+4\,l^{8}m^{2}z^{4}+12\,l^{8}mz^{5}+3\,l^{8}z^{6}-8\,l^{6}m^{2}z^{6}+3\,l^{6}z^{8}-8\,l^{4}m^{2}z^{8}-12\,l^{4}mz^{9}-3\,l^{4}z^{10}+4\,l^{2}m^{2}z^{10}-l^{2}z^{12}+4\,m^{2}z^{12}+4\,mz^{13}+z^{14}} \\
& - \frac{4\,l^{10}mz^{3}\frac{\partial}{\partial v}H^{2}}{l^{14}-4\,l^{12}mz-l^{12}z^{2}+4\,l^{10}m^{2}z^{2}-3\,l^{10}z^{4}+4\,l^{8}m^{2}z^{4}+12\,l^{8}mz^{5}+3\,l^{8}z^{6}-8\,l^{6}m^{2}z^{6}+3\,l^{6}z^{8}-8\,l^{4}m^{2}z^{8}-12\,l^{4}mz^{9}-3\,l^{4}z^{10}+4\,l^{2}m^{2}z^{10}-l^{2}z^{12}+4\,m^{2}z^{12}+4\,mz^{13}+z^{14}} \\
& + \frac{l^{10}z^{4}\frac{\partial}{\partial v}H^{2}}{4\,(l^{14}-4\,l^{12}mz-l^{12}z^{2}+4\,l^{10}m^{2}z^{2}-3\,l^{10}z^{4}+4\,l^{8}m^{2}z^{4}+12\,l^{8}mz^{5}+3\,l^{8}z^{6}-8\,l^{6}m^{2}z^{6}+3\,l^{6}z^{8}-8\,l^{4}m^{2}z^{8}-12\,l^{4}mz^{9}-3\,l^{4}z^{10}+4\,l^{2}m^{2}z^{10}-l^{2}z^{12}+4\,m^{2}z^{12}+4\,mz^{13}+z^{14})} \\
& + \frac{5\,l^{8}m^{2}z^{4}\frac{\partial}{\partial v}H^{2}}{l^{14}-4\,l^{12}mz-l^{12}z^{2}+4\,l^{10}m^{2}z^{2}-3\,l^{10}z^{4}+4\,l^{8}m^{2}z^{4}+12\,l^{8}mz^{5}+3\,l^{8}z^{6}-8\,l^{6}m^{2}z^{6}+3\,l^{6}z^{8}-8\,l^{4}m^{2}z^{8}-12\,l^{4}mz^{9}-3\,l^{4}z^{10}+4\,l^{2}m^{2}z^{10}-l^{2}z^{12}+4\,m^{2}z^{12}+4\,mz^{13}+z^{14}} \\
& - \frac{5\,l^{8}mz^{5}\frac{\partial}{\partial v}H^{2}}{l^{14}-4\,l^{12}mz-l^{12}z^{2}+4\,l^{10}m^{2}z^{2}-3\,l^{10}z^{4}+4\,l^{8}m^{2}z^{4}+12\,l^{8}mz^{5}+3\,l^{8}z^{6}-8\,l^{6}m^{2}z^{6}+3\,l^{6}z^{8}-8\,l^{4}m^{2}z^{8}-12\,l^{4}mz^{9}-3\,l^{4}z^{10}+4\,l^{2}m^{2}z^{10}-l^{2}z^{12}+4\,m^{2}z^{12}+4\,mz^{13}+z^{14}} \\
& - \frac{5\,l^{8}z^{6}\frac{\partial}{\partial v}H^{2}}{4\,(l^{14}-4\,l^{12}mz-l^{12}z^{2}+4\,l^{10}m^{2}z^{2}-3\,l^{10}z^{4}+4\,l^{8}m^{2}z^{4}+12\,l^{8}mz^{5}+3\,l^{8}z^{6}-8\,l^{6}m^{2}z^{6}+3\,l^{6}z^{8}-8\,l^{4}m^{2}z^{8}-12\,l^{4}mz^{9}-3\,l^{4}z^{10}+4\,l^{2}m^{2}z^{10}-l^{2}z^{12}+4\,m^{2}z^{12}+4\,mz^{13}+z^{14})} \\
& + \frac{10\,l^{6}m^{2}z^{6}\frac{\partial}{\partial v}H^{2}}{l^{14}-4\,l^{12}mz-l^{12}z^{2}+4\,l^{10}m^{2}z^{2}-3\,l^{10}z^{4}+4\,l^{8}m^{2}z^{4}+12\,l^{8}mz^{5}+3\,l^{8}z^{6}-8\,l^{6}m^{2}z^{6}+3\,l^{6}z^{8}-8\,l^{4}m^{2}z^{8}-12\,l^{4}mz^{9}-3\,l^{4}z^{10}+4\,l^{2}m^{2}z^{10}-l^{2}z^{12}+4\,m^{2}z^{12}+4\,mz^{13}+z^{14}} \\
& - \frac{5\,l^{6}z^{8}\frac{\partial}{\partial v}H^{2}}{4\,(l^{14}-4\,l^{12}mz-l^{12}z^{2}+4\,l^{10}m^{2}z^{2}-3\,l^{10}z^{4}+4\,l^{8}m^{2}z^{4}+12\,l^{8}mz^{5}+3\,l^{8}z^{6}-8\,l^{6}m^{2}z^{6}+3\,l^{6}z^{8}-8\,l^{4}m^{2}z^{8}-12\,l^{4}mz^{9}-3\,l^{4}z^{10}+4\,l^{2}m^{2}z^{10}-l^{2}z^{12}+4\,m^{2}z^{12}+4\,mz^{13}+z^{14})} + \dots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{12} = &-\frac{\sqrt{2}l^{16}H\frac{\partial^2}{\partial z\partial v}H}{4\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&+\frac{\sqrt{2}l^{14}mzH\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&-\frac{\sqrt{2}l^{14}z^2H\frac{\partial^2}{\partial z\partial v}H}{l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}}\\
&+\frac{5\,\sqrt{2}l^{12}mz^3H\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&-\frac{\sqrt{2}l^{12}z^4H\frac{\partial^2}{\partial z\partial v}H}{l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}}\\
&+\frac{9\,\sqrt{2}l^{10}mz^5H\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&+\frac{\sqrt{2}l^{10}z^6H\frac{\partial^2}{\partial z\partial v}H}{l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}}\\
&+\frac{5\,\sqrt{2}l^8mz^7H\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&+\frac{5\,\sqrt{2}l^8z^8H\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&-\frac{5\,\sqrt{2}l^6mz^9H\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&+\frac{\sqrt{2}l^6z^{10}H\frac{\partial^2}{\partial z\partial v}H}{l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}}\\
&-\frac{9\,\sqrt{2}l^4mz^{11}H\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})} - \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{20} = {} & \frac{5\,l^{10}}{8\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} - \frac{l^{8}mz}{4\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} \\
& - \frac{11\,l^{8}z^{2}}{8\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} - \frac{11\,l^{6}m^{2}z^{2}}{2\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} \\
& - \frac{23\,l^{6}mz^{3}}{4\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} + \frac{4\,l^{4}m^{3}z^{3}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} \\
& - \frac{l^{6}z^{4}}{2\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} + \frac{7\,l^{4}m^{2}z^{4}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} \\
& - \frac{2\,l^{2}m^{4}z^{4}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} + \frac{5\,l^{4}mz^{5}}{4\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} \\
& - \frac{3\,l^{2}m^{3}z^{5}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} + \frac{5\,l^{4}z^{6}}{2\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} \\
& - \frac{l^{2}m^{2}z^{6}}{2\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} - \frac{9\,l^{2}mz^{7}}{4\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} \\
& + \frac{m^{3}z^{7}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} - \frac{9\,l^{2}z^{8}}{8\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} \\
& - \frac{m^{2}z^{8}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} - \frac{mz^{9}}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} \\
& - \frac{z^{10}}{8\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} + \frac{i\,\sqrt{l^{2}+z^{2}}\sqrt{-l^{2}+2\,mz+z^{2}}l^{7}m}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} \\
& + \frac{9i\,\sqrt{l^{2}+z^{2}}\sqrt{-l^{2}+2\,mz+z^{2}}l^{7}z}{4\,(l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12})} - \frac{2i\,\sqrt{l^{2}+z^{2}}\sqrt{-l^{2}+2\,mz+z^{2}}l^{5}m^{2}z}{l^{12}-2\,l^{10}mz-2\,l^{8}mz^{3}-3\,l^{8}z^{4}+4\,l^{6}mz^{5}+4\,l^{4}mz^{7}+3\,l^{4}z^{8}-2\,l^{2}mz^{9}-2\,mz^{11}-z^{12}} - \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{21} = &-\frac{\sqrt{2}l^{16}H\frac{\partial^2}{\partial z\partial v}H}{4\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&+\frac{\sqrt{2}l^{14}mzH\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&-\frac{\sqrt{2}l^{14}z^2H\frac{\partial^2}{\partial z\partial v}H}{l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}}\\
&+\frac{5\,\sqrt{2}l^{12}mz^3H\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&-\frac{\sqrt{2}l^{12}z^4H\frac{\partial^2}{\partial z\partial v}H}{l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}}\\
&+\frac{9\,\sqrt{2}l^{10}mz^5H\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&+\frac{\sqrt{2}l^{10}z^6H\frac{\partial^2}{\partial z\partial v}H}{l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}}\\
&+\frac{5\,\sqrt{2}l^8mz^7H\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&+\frac{5\,\sqrt{2}l^8z^8H\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&-\frac{5\,\sqrt{2}l^6mz^9H\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})}\\
&+\frac{\sqrt{2}l^6z^{10}H\frac{\partial^2}{\partial z\partial v}H}{l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}}\\
&-\frac{9\,\sqrt{2}l^4mz^{11}H\frac{\partial^2}{\partial z\partial v}H}{2\,(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16})} - \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{22} = {} & \frac{l^{20}\frac{\partial}{\partial z}H^2}{2\,(l^{20}-2\,l^{18}mz+4\,l^{18}z^2-10\,l^{16}mz^3+3\,l^{16}z^4-16\,l^{14}mz^5-8\,l^{14}z^6-14\,l^{12}z^8+28\,l^{10}mz^9+28\,l^{8}mz^{11}+14\,l^{8}z^{12}+8\,l^{6}z^{14}-16\,l^{4}mz^{15}-3\,l^{4}z^{16}-10\,l^{2}mz^{17}-4\,l^{2}z^{18}-2\,mz^{19}-z^{20})} \\
& - \frac{l^{18}mz\frac{\partial}{\partial z}H^2}{l^{20}-2\,l^{18}mz+4\,l^{18}z^2-10\,l^{16}mz^3+3\,l^{16}z^4-16\,l^{14}mz^5-8\,l^{14}z^6-14\,l^{12}z^8+28\,l^{10}mz^9+28\,l^{8}mz^{11}+14\,l^{8}z^{12}+8\,l^{6}z^{14}-16\,l^{4}mz^{15}-3\,l^{4}z^{16}-10\,l^{2}mz^{17}-4\,l^{2}z^{18}-2\,mz^{19}-z^{20}} \\
& + \frac{2\,l^{18}z^2\frac{\partial}{\partial z}H^2}{l^{20}-2\,l^{18}mz+4\,l^{18}z^2-10\,l^{16}mz^3+3\,l^{16}z^4-16\,l^{14}mz^5-8\,l^{14}z^6-14\,l^{12}z^8+28\,l^{10}mz^9+28\,l^{8}mz^{11}+14\,l^{8}z^{12}+8\,l^{6}z^{14}-16\,l^{4}mz^{15}-3\,l^{4}z^{16}-10\,l^{2}mz^{17}-4\,l^{2}z^{18}-2\,mz^{19}-z^{20}} \\
& - \frac{5\,l^{16}mz^3\frac{\partial}{\partial z}H^2}{l^{20}-2\,l^{18}mz+4\,l^{18}z^2-10\,l^{16}mz^3+3\,l^{16}z^4-16\,l^{14}mz^5-8\,l^{14}z^6-14\,l^{12}z^8+28\,l^{10}mz^9+28\,l^{8}mz^{11}+14\,l^{8}z^{12}+8\,l^{6}z^{14}-16\,l^{4}mz^{15}-3\,l^{4}z^{16}-10\,l^{2}mz^{17}-4\,l^{2}z^{18}-2\,mz^{19}-z^{20}} \\
& + \frac{3\,l^{16}z^4\frac{\partial}{\partial z}H^2}{2\,(l^{20}-2\,l^{18}mz+4\,l^{18}z^2-10\,l^{16}mz^3+3\,l^{16}z^4-16\,l^{14}mz^5-8\,l^{14}z^6-14\,l^{12}z^8+28\,l^{10}mz^9+28\,l^{8}mz^{11}+14\,l^{8}z^{12}+8\,l^{6}z^{14}-16\,l^{4}mz^{15}-3\,l^{4}z^{16}-10\,l^{2}mz^{17}-4\,l^{2}z^{18}-2\,mz^{19}-z^{20})} \\
& - \frac{8\,l^{14}mz^5\frac{\partial}{\partial z}H^2}{l^{20}-2\,l^{18}mz+4\,l^{18}z^2-10\,l^{16}mz^3+3\,l^{16}z^4-16\,l^{14}mz^5-8\,l^{14}z^6-14\,l^{12}z^8+28\,l^{10}mz^9+28\,l^{8}mz^{11}+14\,l^{8}z^{12}+8\,l^{6}z^{14}-16\,l^{4}mz^{15}-3\,l^{4}z^{16}-10\,l^{2}mz^{17}-4\,l^{2}z^{18}-2\,mz^{19}-z^{20}} \\
& - \frac{4\,l^{14}z^6\frac{\partial}{\partial z}H^2}{l^{20}-2\,l^{18}mz+4\,l^{18}z^2-10\,l^{16}mz^3+3\,l^{16}z^4-16\,l^{14}mz^5-8\,l^{14}z^6-14\,l^{12}z^8+28\,l^{10}mz^9+28\,l^{8}mz^{11}+14\,l^{8}z^{12}+8\,l^{6}z^{14}-16\,l^{4}mz^{15}-3\,l^{4}z^{16}-10\,l^{2}mz^{17}-4\,l^{2}z^{18}-2\,mz^{19}-z^{20}} \\
& - \frac{7\,l^{12}z^8\frac{\partial}{\partial z}H^2}{l^{20}-2\,l^{18}mz+4\,l^{18}z^2-10\,l^{16}mz^3+3\,l^{16}z^4-16\,l^{14}mz^5-8\,l^{14}z^6-14\,l^{12}z^8+28\,l^{10}mz^9+28\,l^{8}mz^{11}+14\,l^{8}z^{12}+8\,l^{6}z^{14}-16\,l^{4}mz^{15}-3\,l^{4}z^{16}-10\,l^{2}mz^{17}-4\,l^{2}z^{18}-2\,mz^{19}-z^{20}} \\
& + \frac{14\,l^{10}mz^9\frac{\partial}{\partial z}H^2}{l^{20}-2\,l^{18}mz+4\,l^{18}z^2-10\,l^{16}mz^3+3\,l^{16}z^4-16\,l^{14}mz^5-8\,l^{14}z^6-14\,l^{12}z^8+28\,l^{10}mz^9+28\,l^{8}mz^{11}+14\,l^{8}z^{12}+8\,l^{6}z^{14}-16\,l^{4}mz^{15}-3\,l^{4}z^{16}-10\,l^{2}mz^{17}-4\,l^{2}z^{18}-2\,mz^{19}-z^{20}} \\
& + \frac{14\,l^{8}mz^{11}\frac{\partial}{\partial z}H^2}{l^{20}-2\,l^{18}mz+4\,l^{18}z^2-10\,l^{16}mz^3+3\,l^{16}z^4-16\,l^{14}mz^5-8\,l^{14}z^6-14\,l^{12}z^8+28\,l^{10}mz^9+28\,l^{8}mz^{11}+14\,l^{8}z^{12}+8\,l^{6}z^{14}-16\,l^{4}mz^{15}-3\,l^{4}z^{16}-10\,l^{2}mz^{17}-4\,l^{2}z^{18}-2\,mz^{19}-z^{20}} \\
& + \frac{7\,l^{8}z^{12}\frac{\partial}{\partial z}H^2}{l^{20}-2\,l^{18}mz+4\,l^{18}z^2-10\,l^{16}mz^3+3\,l^{16}z^4-16\,l^{14}mz^5-8\,l^{14}z^6-14\,l^{12}z^8+28\,l^{10}mz^9+28\,l^{8}mz^{11}+14\,l^{8}z^{12}+8\,l^{6}z^{14}-16\,l^{4}mz^{15}-3\,l^{4}z^{16}-10\,l^{2}mz^{17}-4\,l^{2}z^{18}-2\,mz^{19}-z^{20}} \\
& + \frac{4\,l^{6}z^{14}\frac{\partial}{\partial z}H^2}{l^{20}-2\,l^{18}mz+4\,l^{18}z^2-10\,l^{16}mz^3+3\,l^{16}z^4-16\,l^{14}mz^5-8\,l^{14}z^6-14\,l^{12}z^8+28\,l^{10}mz^9+28\,l^{8}mz^{11}+14\,l^{8}z^{12}+8\,l^{6}z^{14}-16\,l^{4}mz^{15}-3\,l^{4}z^{16}-10\,l^{2}mz^{17}-4\,l^{2}z^{18}-2\,mz^{19}-z^{20}} - \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = {} & -\frac{l^{14}\frac{\partial}{\partial v}H^2}{12\,(l^{14}-4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^{8}m^2z^4+12\,l^{8}mz^5+3\,l^{8}z^6-8\,l^{6}m^2z^6+3\,l^{6}z^8-8\,l^{4}m^2z^8-12\,l^{4}mz^9-3\,l^{4}z^{10}+4\,l^{2}m^2z^{10}-l^{2}z^{12}+4\,m^2z^{12}+4\,mz^{13}+z^{14})} \\
& + \frac{l^{12}mz\frac{\partial}{\partial v}H^2}{3\,(l^{14}-4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^{8}m^2z^4+12\,l^{8}mz^5+3\,l^{8}z^6-8\,l^{6}m^2z^6+3\,l^{6}z^8-8\,l^{4}m^2z^8-12\,l^{4}mz^9-3\,l^{4}z^{10}+4\,l^{2}m^2z^{10}-l^{2}z^{12}+4\,m^2z^{12}+4\,mz^{13}+z^{14})} \\
& - \frac{l^{12}z^2\frac{\partial}{\partial v}H^2}{4\,(l^{14}-4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^{8}m^2z^4+12\,l^{8}mz^5+3\,l^{8}z^6-8\,l^{6}m^2z^6+3\,l^{6}z^8-8\,l^{4}m^2z^8-12\,l^{4}mz^9-3\,l^{4}z^{10}+4\,l^{2}m^2z^{10}-l^{2}z^{12}+4\,m^2z^{12}+4\,mz^{13}+z^{14})} \\
& - \frac{l^{10}m^2z^2\frac{\partial}{\partial v}H^2}{3\,(l^{14}-4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^{8}m^2z^4+12\,l^{8}mz^5+3\,l^{8}z^6-8\,l^{6}m^2z^6+3\,l^{6}z^8-8\,l^{4}m^2z^8-12\,l^{4}mz^9-3\,l^{4}z^{10}+4\,l^{2}m^2z^{10}-l^{2}z^{12}+4\,m^2z^{12}+4\,mz^{13}+z^{14})} \\
& + \frac{4\,l^{10}mz^3\frac{\partial}{\partial v}H^2}{3\,(l^{14}-4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^{8}m^2z^4+12\,l^{8}mz^5+3\,l^{8}z^6-8\,l^{6}m^2z^6+3\,l^{6}z^8-8\,l^{4}m^2z^8-12\,l^{4}mz^9-3\,l^{4}z^{10}+4\,l^{2}m^2z^{10}-l^{2}z^{12}+4\,m^2z^{12}+4\,mz^{13}+z^{14})} \\
& - \frac{l^{10}z^4\frac{\partial}{\partial v}H^2}{12\,(l^{14}-4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^{8}m^2z^4+12\,l^{8}mz^5+3\,l^{8}z^6-8\,l^{6}m^2z^6+3\,l^{6}z^8-8\,l^{4}m^2z^8-12\,l^{4}mz^9-3\,l^{4}z^{10}+4\,l^{2}m^2z^{10}-l^{2}z^{12}+4\,m^2z^{12}+4\,mz^{13}+z^{14})} \\
& - \frac{5\,l^{8}m^2z^4\frac{\partial}{\partial v}H^2}{3\,(l^{14}-4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^{8}m^2z^4+12\,l^{8}mz^5+3\,l^{8}z^6-8\,l^{6}m^2z^6+3\,l^{6}z^8-8\,l^{4}m^2z^8-12\,l^{4}mz^9-3\,l^{4}z^{10}+4\,l^{2}m^2z^{10}-l^{2}z^{12}+4\,m^2z^{12}+4\,mz^{13}+z^{14})} \\
& + \frac{5\,l^{8}mz^5\frac{\partial}{\partial v}H^2}{3\,(l^{14}-4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^{8}m^2z^4+12\,l^{8}mz^5+3\,l^{8}z^6-8\,l^{6}m^2z^6+3\,l^{6}z^8-8\,l^{4}m^2z^8-12\,l^{4}mz^9-3\,l^{4}z^{10}+4\,l^{2}m^2z^{10}-l^{2}z^{12}+4\,m^2z^{12}+4\,mz^{13}+z^{14})} \\
& + \frac{5\,l^{8}z^6\frac{\partial}{\partial v}H^2}{12\,(l^{14}-4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^{8}m^2z^4+12\,l^{8}mz^5+3\,l^{8}z^6-8\,l^{6}m^2z^6+3\,l^{6}z^8-8\,l^{4}m^2z^8-12\,l^{4}mz^9-3\,l^{4}z^{10}+4\,l^{2}m^2z^{10}-l^{2}z^{12}+4\,m^2z^{12}+4\,mz^{13}+z^{14})} \\
& - \frac{10\,l^{6}m^2z^6\frac{\partial}{\partial v}H^2}{3\,(l^{14}-4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^{8}m^2z^4+12\,l^{8}mz^5+3\,l^{8}z^6-8\,l^{6}m^2z^6+3\,l^{6}z^8-8\,l^{4}m^2z^8-12\,l^{4}mz^9-3\,l^{4}z^{10}+4\,l^{2}m^2z^{10}-l^{2}z^{12}+4\,m^2z^{12}+4\,mz^{13}+z^{14})} \\
& + \frac{5\,l^{6}z^8\frac{\partial}{\partial v}H^2}{12\,(l^{14}-4\,l^{12}mz-l^{12}z^2+4\,l^{10}m^2z^2-3\,l^{10}z^4+4\,l^{8}m^2z^4+12\,l^{8}mz^5+3\,l^{8}z^6-8\,l^{6}m^2z^6+3\,l^{6}z^8-8\,l^{4}m^2z^8-12\,l^{4}mz^9-3\,l^{4}z^{10}+4\,l^{2}m^2z^{10}-l^{2}z^{12}+4\,m^2z^{12}+4\,mz^{13}+z^{14})} - \cdots
\end{aligned}
$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$$
\begin{aligned}
\Psi_2 = {} & \frac{l^{14}\frac{\partial}{\partial v}H^2}{6\,(l^{14} - 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 + 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 - 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} + 4\,mz^{13} + z^{14})} \\
& - \frac{2\,l^{12}mz\frac{\partial}{\partial v}H^2}{3\,(l^{14} - 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 + 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 - 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} + 4\,mz^{13} + z^{14})} \\
& + \frac{l^{12}z^2\frac{\partial}{\partial v}H^2}{2\,(l^{14} - 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 + 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 - 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} + 4\,mz^{13} + z^{14})} \\
& + \frac{2\,l^{10}m^2z^2\frac{\partial}{\partial v}H^2}{3\,(l^{14} - 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 + 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 - 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} + 4\,mz^{13} + z^{14})} \\
& - \frac{8\,l^{10}mz^3\frac{\partial}{\partial v}H^2}{3\,(l^{14} - 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 + 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 - 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} + 4\,mz^{13} + z^{14})} \\
& + \frac{l^{10}z^4\frac{\partial}{\partial v}H^2}{6\,(l^{14} - 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 + 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 - 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} + 4\,mz^{13} + z^{14})} \\
& + \frac{10\,l^8m^2z^4\frac{\partial}{\partial v}H^2}{3\,(l^{14} - 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 + 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 - 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} + 4\,mz^{13} + z^{14})} \\
& - \frac{10\,l^8mz^5\frac{\partial}{\partial v}H^2}{3\,(l^{14} - 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 + 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 - 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} + 4\,mz^{13} + z^{14})} \\
& - \frac{5\,l^8z^6\frac{\partial}{\partial v}H^2}{6\,(l^{14} - 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 + 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 - 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} + 4\,mz^{13} + z^{14})} \\
& + \frac{20\,l^6m^2z^6\frac{\partial}{\partial v}H^2}{3\,(l^{14} - 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 + 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 - 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} + 4\,mz^{13} + z^{14})} \\
& - \frac{5\,l^6z^8\frac{\partial}{\partial v}H^2}{6\,(l^{14} - 4\,l^{12}mz - l^{12}z^2 + 4\,l^{10}m^2z^2 - 3\,l^{10}z^4 + 4\,l^8m^2z^4 + 12\,l^8mz^5 + 3\,l^8z^6 - 8\,l^6m^2z^6 + 3\,l^6z^8 - 8\,l^4m^2z^8 - 12\,l^4mz^9 - 3\,l^4z^{10} + 4\,l^2m^2z^{10} - l^2z^{12} + 4\,m^2z^{12} + 4\,mz^{13} + z^{14})} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Psi_3 = &-\frac{\sqrt{2}l^{16}H\frac{\partial^2}{\partial z\partial v}H}{4\left(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}\right)}\\
&+\frac{\sqrt{2}l^{14}mzH\frac{\partial^2}{\partial z\partial v}H}{2\left(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}\right)}\\
&-\frac{\sqrt{2}l^{14}z^2H\frac{\partial^2}{\partial z\partial v}H}{l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}}\\
&+\frac{5\,\sqrt{2}l^{12}mz^3H\frac{\partial^2}{\partial z\partial v}H}{2\left(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}\right)}\\
&-\frac{\sqrt{2}l^{12}z^4H\frac{\partial^2}{\partial z\partial v}H}{l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}}\\
&+\frac{9\,\sqrt{2}l^{10}mz^5H\frac{\partial^2}{\partial z\partial v}H}{2\left(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}\right)}\\
&+\frac{\sqrt{2}l^{10}z^6H\frac{\partial^2}{\partial z\partial v}H}{l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}}\\
&+\frac{5\,\sqrt{2}l^8mz^7H\frac{\partial^2}{\partial z\partial v}H}{2\left(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}\right)}\\
&+\frac{5\,\sqrt{2}l^8z^8H\frac{\partial^2}{\partial z\partial v}H}{2\left(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}\right)}\\
&-\frac{5\,\sqrt{2}l^6mz^9H\frac{\partial^2}{\partial z\partial v}H}{2\left(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}\right)}\\
&+\frac{\sqrt{2}l^6z^{10}H\frac{\partial^2}{\partial z\partial v}H}{l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}}\\
&-\frac{9\,\sqrt{2}l^4mz^{11}H\frac{\partial^2}{\partial z\partial v}H}{2\left(l^{16}-2\,l^{14}mz+2\,l^{14}z^2-6\,l^{12}mz^3-2\,l^{12}z^4-2\,l^{10}mz^5-6\,l^{10}z^6+10\,l^8mz^7+10\,l^6mz^9+6\,l^6z^{10}-2\,l^4mz^{11}+2\,l^4z^{12}-6\,l^2mz^{13}-2\,l^2z^{14}-2\,mz^{15}-z^{16}\right)}-\cdots
\end{aligned}
$$

$$
\begin{aligned}
\Psi_4 = {} & \frac{l^{12}\frac{\partial}{\partial z}H^2}{2\,(l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12})} - \frac{l^{10}mz\frac{\partial}{\partial z}H^2}{l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12}} \\
& - \frac{l^8mz^3\frac{\partial}{\partial z}H^2}{l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12}} - \frac{3\,l^8z^4\frac{\partial}{\partial z}H^2}{2\,(l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12})} \\
& + \frac{2\,l^6mz^5\frac{\partial}{\partial z}H^2}{l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12}} + \frac{2\,l^4mz^7\frac{\partial}{\partial z}H^2}{l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12}} \\
& + \frac{3\,l^4z^8\frac{\partial}{\partial z}H^2}{2\,(l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12})} - \frac{l^2mz^9\frac{\partial}{\partial z}H^2}{l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12}} \\
& - \frac{mz^{11}\frac{\partial}{\partial z}H^2}{l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12}} - \frac{z^{12}\frac{\partial}{\partial z}H^2}{2\,(l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12})} \\
& + \frac{l^{12}H\frac{\partial^2}{(\partial z)^2}H}{2\,(l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12})} - \frac{l^{10}mzH\frac{\partial^2}{(\partial z)^2}H}{l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12}} \\
& - \frac{l^8mz^3H\frac{\partial^2}{(\partial z)^2}H}{l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12}} - \frac{3\,l^8z^4H\frac{\partial^2}{(\partial z)^2}H}{2\,(l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12})} \\
& + \frac{2\,l^6mz^5H\frac{\partial^2}{(\partial z)^2}H}{l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12}} + \frac{2\,l^4mz^7H\frac{\partial^2}{(\partial z)^2}H}{l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12}} \\
& + \frac{3\,l^4z^8H\frac{\partial^2}{(\partial z)^2}H}{2\,(l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12})} - \frac{l^2mz^9H\frac{\partial^2}{(\partial z)^2}H}{l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12}} \\
& - \frac{mz^{11}H\frac{\partial^2}{(\partial z)^2}H}{l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12}} - \frac{z^{12}H\frac{\partial^2}{(\partial z)^2}H}{2\,(l^{12}-2\,l^{10}mz-2\,l^8mz^3-3\,l^8z^4+4\,l^6mz^5+4\,l^4mz^7+3\,l^4z^8-2\,l^2mz^9-2\,mz^{11}-z^{12})} + \dots
\end{aligned}
$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"